Technical report — Swiss Heritage Factuality: does Apertus recognise Switzerland's landmarks?

Final labels are the frozen LLM judge's (`judge-v1`; multiple choice: code), checked on 522 Apertus-70B answers by one blind human annotator (judge–human κ = 0.76). Photos were screened by AI (see Annotation and Limitations).

We built a four-language (FR/DE/IT/RM) benchmark of 5,420 questions on 251 Swiss heritage landmarks from the federal inventory (KGS), stratified by language region and fame, with photo, second-view, multiple-choice, text-only and trap variants and per-item licences. Apertus 1.5 70B almost never recognises a landmark from its photo: given four named options, it is right only 35 % of the time (chance 25 %; the open model Gemma-4-31B: 61 %), about equally in all four languages. The language changes whether it admits this. On the same photos it mostly declines in French (11 % of names confidently wrong) but answers in Romansh (68 % of names and 97 % of places wrong). In Romansh it also moves half of all non-Graubünden landmarks to Graubünden, a pull Gemma shares. Shown a too-generic photo, it declines every time; Gemma invents a name for a quarter.

Track: Apertus Readiness - Track 1B (Swiss Voices), dimension: core-task intelligence (factuality)
Event: Online

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Demo / data: dataset `huggingface.co/datasets/chrizzto/swiss-heritage-factuality` · code `github.com/ChristoYusj/aper swiss-landmarks`

Use of Apertus

- **Model:** `swiss-ai/Apertus-v1.5-70B` (main), `swiss-ai/Apertus-v1.5-8B` (secondary); comparison model `google/gemma-4-31B-it` (open weights, via the Hugging Face router) on the same items
- **How it is used:** evaluation (multimodal: photo + question)
- **Where it runs:** hosted CSCS inference endpoint (OpenAI-compatible); the harness accepts any OpenAI-compatible endpoint via `LLM_NAME`, `LLM_BASE_URL`, `LLM_API_KEY`

Use Case Description

A visitor or a teacher points a phone at a Swiss chapel, castle or fountain and asks "what is this, where is it, when was it built?". This is a common use of a Swiss multimodal assistant or tourism chatbot. A confident wrong answer is costly here: it is believable, about a real place, and the asker usually cannot check it. Switzerland's heritage is also a clean test of *localised* knowledge: the federal inventory of cultural property (KGS) lists thousands of objects, most known only locally, across four language regions of very unequal size.

We measure the **factuality and hallucination** of Apertus 1.5 on these landmarks (SimpleQA-style: correct / incorrect / not attempted). Four questions were fixed before the run (pre-registration commit `a73986e`, `docs/analysis_plan.md`): Does Apertus invent more about obscure landmarks (H1)? Are wrong answers pulled toward the German-speaking majority (H2) or toward famous look-alikes (H2b)? Is the failure one of *recognition* (photo) rather than *knowledge* (name given) (H3)? Does it accept a false premise in the question (H4)?

Dataset Summary

Landmarks	251 KGS heritage objects (A: 104, B: 139, local: 8) in 4 BFS language regions × 3 fame tiers (264 drawn; 13 removed by the photo screening)
Fame tiers	Wikipedia articles: low 0, mid 1–2, high ≥ 3 (21–24 per cell; Romansh region: smaller pools, 12–18, the high-fame cell a census)
Types	churches 108, castles 61, museums 30, civic 18, bridges 14, fountains 10, towers/gates 8, monuments 2

Photos	251 main photos (Wikidata P18), 161 second views, 17 too-generic photos; 500 px, EXIF stripped, per-image licence
Questions	5,420: core 3,412 (`photo_main` 2,276, each photo in 4 languages; `photo_alt` 732; `text_only` 322; `trap_premise` 48; `trap_lowinfo` 34) + registered extensions 2,008 (`photo_mc` 1,004; `photo_main_w2` 1,004)
Question types	Q1 name, Q2 municipality + canton, Q3 year built (67 landmarks with a precise, unconflicted year)
Languages	French, German, Italian, Romansh (Rumantsch Grischun); identical templates, explicit "if unsure, say so"
Answer keys	Wikidata (CC0) + BFS municipality register: name aliases (flagged if only valid with the town), current/former municipality names, canton names in 5 languages, year windows
Licences	questions/keys/responses CDLA-Permissive-2.0; photos keep their CC licence (TASL attribution per item)

Arms. `photo_main`: the photo, asked in all four languages (same image, so language is the only change). `text_only` (FR, DE): the name is given and the place or year asked, separating *recognition* from *knowledge*. `trap_premise` (false-premise trap): the question asserts the town of a real look-alike in another municipality ("This building is in Rothrist. Which building is it?"). `photo_alt` (FR, DE): a second view of the same object, an indirect memorisation check. `trap_lowinfo` (FR, DE): a too-generic photo; declining is right. Extensions, registered before they were run (`docs/extensions_plan.md`): `photo_mc` (multiple choice): the main photo with four named options (the landmark and three others of the same type, mostly from the same region; chance 25 %), in all four languages; `photo_main_w2` (second wording; FR, RM): the name and place questions, independently reworded.

Selection funnel (seed 20261013): 13,417 heritage objects with a photo → 4,153 eligible (public landmark types, a current municipality, a specific label, one language region) → stratified sample of 264 + 83 same-cell reserves for replacements.

Data Collection Method

Sources. Wikidata (QLever mirror, cached queries in the repository) for objects, names, inception dates and administrative location; the Federal Statistical Office municipality register (01-10-2026) for municipality numbers, cantons and language regions; Wikimedia Commons for photos and their licences.

Answer keys (rules in code, decisions in `CHANGELOG.md`). Names: Wikidata labels, aliases and sitelinks in six languages, minus town-only and type-only aliases. A name is *non-distinctive* if nothing remains after removing type words, saints and the town ("Kapelle St. Antonius"); it then counts only with the right town. Places: the current municipality's names in all languages (16 district/region/former-municipality labels removed); shared town names need the canton. Years: unconflicted year- or decade-precision inception (window ±5), never a museum's founding date. An independent script re-checked 25 random landmarks against live Wikidata (25/25).

Question validity rules (from the question text only): text-only questions are dropped when the given name is non-distinctive or contains the town or a year (170 items, kept with their answers in `metadata/excluded_items.jsonl`).

Annotation (`docs/annotation_guidelines.md`; deviations from the plan are logged in `CHANGELOG.md`).

- *Photo screening, by AI (deviation: the plan had the team screen).* Rules: exclude recognisable people, readable plates, private interiors and personal text (never blur); drop a main photo that does not show the object; pick a second view; mark too-generic photos. Claude Opus 5.5 screened all 917 candidates, three further instances re-checked the 491 proposed for shipping, and a second AI agent (Hermes) reviewed them under the team lead's "if unsure, exclude" policy; the team lead decided 6 edge cases. 13 landmarks were removed (privacy 4, wrong object 4, unclear 5; answers kept in `metadata/excluded_items.jsonl`); 429 photos shipped. [P1]
- *Grading audit (human, blind):* one annotator (A1, the team lead) graded the pre-registered, seeded, stratified 20 % sample (546 of the 2,726 70B answers before the screening; 522 still in the dataset; not covered: `photo_alt`, `trap_lowinfo`, the extensions, 8B, Gemma) on three fillable PDFs that show no grade of any kind, with an English machine gloss for non-French answers. The protocol planned three annotators; A1 graded all three parts, so no agreement between people is reported. Hermes graded the same sample as an AI cross-check.

Consent and personal data. No person is a data subject: photos with recognisable people are excluded and EXIF is stripped. Annotator consent form: `docs/consent_form_template.md` (pseudonymous IDs).

Dataset Details

- **Quality metrics:** answer-key audit (above), selection funnel, per-cell counts, question validity rules, stability of answers at temperature 0 (below), judge–code agreement, an AI-assisted re-read of every core Apertus disagreement, a blind human audit of 522 70B answers (one annotator) and an AI cross-check.
- **Diversity / edge cases:** four languages incl. Romansh; fame tiers; same-photo language contrast; text vs photo; false-premise traps with real look-alikes; too-generic photos; ambiguous town names; non-distinctive names.
- **Contamination:** the Apertus pre-training data list names no Wikimedia or landmark image set, but includes web-scale image–text corpora that may contain some of these photos; we cannot check at URL level (limitation). Our indirect check, the `photo_main` vs `photo_alt` contrast, shows no advantage for the main photos (Results 4).
- **Ethics and licensing:** photos with NC/ND/GFDL-only/unknown licences are excluded; every image row carries author, licence, source URL and the modification made. Takedown requests: the dataset's discussion page.

Evaluation

Method

- **Labels:** SimpleQA three-way. *Hallucination rate* = incorrect / attempted.
- **Graders:** (1) a deterministic code grader on 100 % of items: alias matching with the distinctiveness and ambiguity rules, abstention and hedge detection in all four languages, and a gazetteer of all Swiss cantons and current and former municipalities. (2) An independent open-weights judge, `openai/gpt-oss-120b` (not Apertus), on every free answer, pinned to one provider, temperature 0. Its prompt was tuned on a 40-item dev set (labels proposed by Claude; the team lead ruled on the 4 hard cases and accepted the other 36 without item-by-item review); the final prompt reached 40/40 agreement, two runs identical, and was frozen before the full run and the audit sample. Final label = the judge's (for 4 empty Gemma verdicts: the code grader's). Multiple choice is graded by code only (one option named by letter or text).
- **Statistics:** cluster bootstrap over landmarks (2,000 resamples) for CIs; exact McNemar for paired contrasts; Holm correction across the registered tests; Wilson intervals for rates.

Evaluation setup and inference parameters

Temperature 0, top_p 1, max_tokens 300, seed 1234, one image (base64 JPEG) + one question, no system prompt. Every response is stored with a content hash of its request; `verify` replays the stored answers. **Stability:** 100 items were re-asked once and a random 20 of them twice more. On the first re-ask, 75 % of the 100 answers had identical text and **99 % an identical code label** (κ = 0.98; the one change: a vague hedge that, asked again, named two candidate cantons). All 20 probe items kept their label over four answers. The endpoint is not bit-for-bit deterministic, but the grades are stable.

Results

1. The question's language decides whether the model answers, not what it recognises.

photo_main, Apertus 70B (n per cell = 251)	FR	IT	DE	RM
Name: not attempted (declines, or too vague to check)	89 %	76 %	58 %	**32 %**
Name: confidently wrong	11 %	24 %	41 %	**68 %**
Place: confidently wrong	2 %	2 %	30 %	**97 %**

Only 4 of 1,004 names and 6 of 1,004 places are right; 360 of 364 committed names are wrong. The same instruction ("if you are not sure, say so") gives honest refusals in French and confident fabrication in Romansh, on identical images. **With**

multiple choice (photo_mc, four named options) the language gap disappears: 70B picks the right landmark in 32–39 % in every language (34.7 % overall, 95 % CI 29.7–39.7 resampling landmarks; chance 25 %), 8B in 31–33 %. Part of the margin over chance is a position bias: 70B starts with "A" in 58 % of answers and is right in 74 % when the correct option is A, about 22 % otherwise (still above the ~14 % a pure "A" bias would give). So recognition is weak in every language. **The second wording** (photo_main_w2, registered before the run for 70B, also run on 8B and Gemma) keeps the gap: 70B gets 93 % of Romansh names wrong vs 20 % of French. The 8B model is very sensitive to wording (French names wrong 78 % → 32 %), but Romansh stays its worst language (79 %). Gemma-4-31B has no such gap: it rarely declines (names not attempted 5–10 %), gives a wrong name in 86–94 % of answers in every language, and in multiple choice is right 61 % of the time (95 % CI 55.8–66.4; 61–63 % in every language). It recognises far more than Apertus but does not say when it does not know. The language-dependent caution is an Apertus trait (both sizes, strongest in 70B).

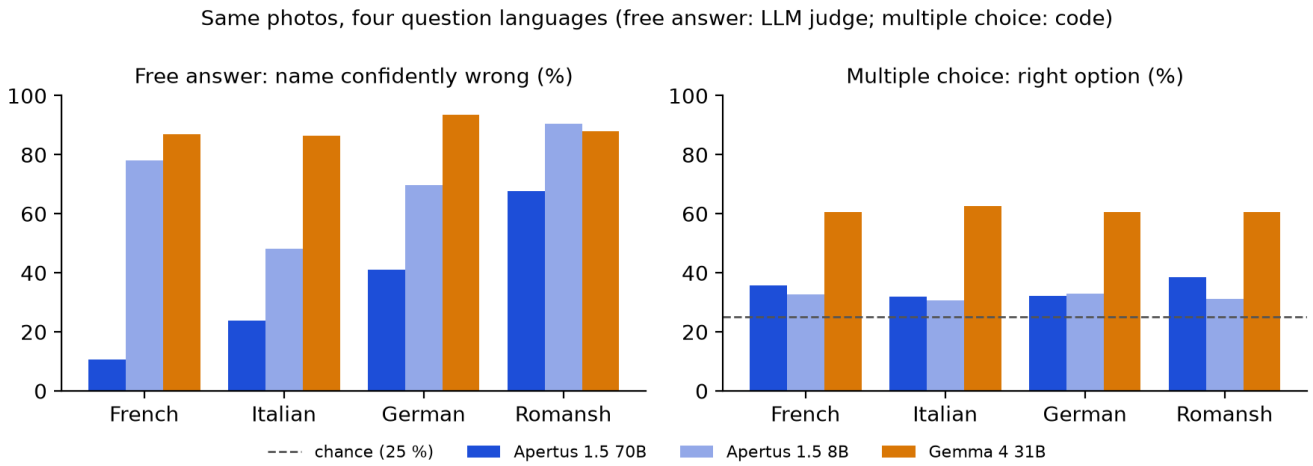


Figure 1: Free answer (left) vs multiple choice (right), same photos, by question language and model.

2. In Romansh the model moves landmarks to Graubünden (exploratory, location extracted with the gazetteer). Of 190 landmarks outside Graubünden, the Romansh answers place **96** in Graubünden; the French and Italian answers for the same photos place none there, the German answers 14 (paired McNemar RM vs FR/IT/DE, $p < 10^{-6}$). Overall, 60 % of Romansh place answers say Graubünden (true share 24 %), mostly naming German- and Italian-speaking towns (93 of 137 named; e.g. Chur, Roveredo, Poschiavo). The pull is toward the canton associated with Romansh, not toward Romansh-speaking villages. The 8B model shows the same pull, more weakly (35 %). Gemma shows it too (58 % of its Romansh place answers say Graubünden; 95 of the 190 non-Graubünden landmarks), so this is a cross-model association of Romansh with Graubünden, not an Apertus quirk.

3. Famous names absorb unknown objects. Eleven different landmarks, mostly castles, are called "Castel Tarasp" (the famous Engadine castle) and eight "Castle Vaduz" (Liechtenstein); in French and Italian answers, castles become "Château de Chillon". Disentis Abbey becomes "Kloster Einsiedeln".

4. A second view and a too-generic photo (descriptive, FR + DE). On the 161 landmarks with a second view, 70B names 2 landmarks correctly from the main photo and 1 other (in both languages) from the second view. It does not do better on the main photo (wrong, second view vs main photo: names 24 % vs 26 %; places 11 % vs 16 %, McNemar $p = 0.04$): no advantage for the Wikidata main photos, so no sign that they were memorised. On 17 too-generic photos, 70B declined all 34 times and 8B 33 of 34, although on the main photos of the same landmarks they gave wrong names 7 and 26 times. The models can tell a too-generic photo from a recognisable one, but a photo that *looks* identifiable triggers a confident guess.

5. Registered hypotheses (Holm-adjusted):

	Result
H1 fame cliff (name)	not supported: hallucination 97.5 % (low) vs 99.2 % (high); floor effect
H1b fame cliff (place)	not supported (Δ = +2.3 pp, CI −1.0 to +6.4)

	Result
H2 majority-region pull	not supported: wrong towns for non-DE landmarks are 59 % German-speaking vs 65 % of all municipalities (registered comparison points the other way, p_Holm = 0.10)
H2b famous look-alike	not testable: 3 answers (1 landmark) name a look-alike's town
H3 seeing vs knowing (place)	**supported in French** (text 8 vs photo 0 correct of 95 landmarks, p_Holm = 0.04), same direction in German (9 vs 1, p_Holm = 0.09). Fragile: before the photo screening removed 13 landmarks, French was 8 vs 1 (p_Holm = 0.13)
H4 false premise	**supported**: 79 % adopt the false town vs 29 % incorrect on the same photo asked plainly (p_Holm < 0.0001); counting a wrongly named building as incorrect too (team rule), 88 %

Knowing the name helps, and there fame matters: with the name given, place accuracy is 12.5 % for high-fame vs 3.7 % for low-fame landmarks; from the photo, both are ≈ 0 %. The bottleneck is visual recognition.

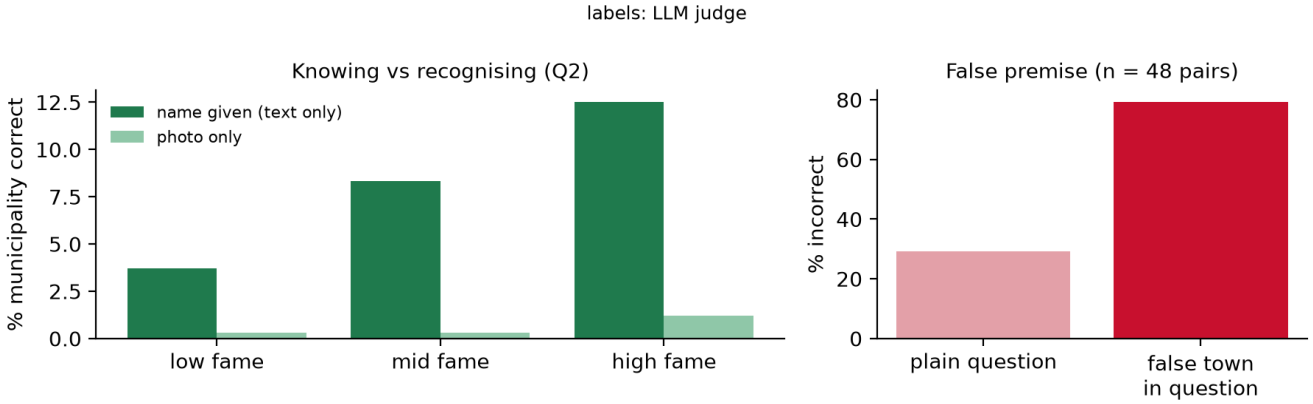


Figure 2: Left: municipality correct with the name given vs from the photo, by fame (H3). Right: incorrect on the plain photo question vs with a false town in the question (H4).

6. Smaller model and comparison model. Apertus 1.5 8B declines less (names not attempted: 28 % vs 64 % for 70B) and is confidently wrong more often in every language (names wrong: FR 78 %, IT 48 %, DE 70 %, RM 90 %). The false premise does not mislead it more than the plain question (71 % vs 83 %, n.s.), because it is already wrong on the plain question. Gemma-4-31B (names: 9 % not attempted, 89 % wrong) names the landmark in 26 of 1,004 answers (13 landmarks; Apertus 70B: 4 answers, 3 landmarks) and is the only model with a significant fame effect (H1: 99.7 % vs 93.2 % hallucination, low vs high fame; H1b likewise; p_Holm 0.01–0.02). It invents a name for 24 % of the too-generic photos (Apertus 70B: 0 %) and adopts the false town in 69 % of false-premise traps.

7. Agreement and judge errors. Judge vs code grader on the 3,412 core items: κ = 0.95 (70B; 97.9 % identical labels), 0.94 (8B; 96.8 %), 0.92 (Gemma; 97.2 %). Every core disagreement of the Apertus models was re-read independently against the written rules, with a second blind read where the judge looked wrong (236 re-reads over two rounds, 224 still in the dataset; `disagreement_adjudication.jsonl`; AI-assisted, not a human audit). Of the 224, the judge was right in 86, the code grader in 131, neither in 1, and 6 stayed unsettled. Second-wording and Gemma disagreements were not re-read. The judge is too strict on vague answers whose building type fits the landmark and on saint names without a town, and labels vague centuries outside the year window as not attempted. Replacing its label wherever the re-read settled it lowers the Romansh name-wrong rate from 68 % to 59 % (70B; 8B 90 % → 81 %), raises the small Romansh year cell by 6 pp and lowers the false-premise trap rate from 88 % to 83 %; no other 70B language × question cell moves by more than about 1.5 pp. The registered results keep the frozen judge's labels. *Human audit* (blind, one annotator, 522 answers): judge vs human κ = 0.76 (89 % identical), code grader vs human κ = 0.73 (88 %). Most disagreements are answers the judge counted as declines but the annotator marked incorrect (29), and answers the annotator marked correct but the judge did not (18). The 30 answers

that appear in all three parts got the same label each time. *AI cross-check:* a second AI agent graded the same sample: κ = 0.96 with the judge (98.2 % identical), 0.99 with the code grader.

Dataset Limitations

- Results use one template per language (plus a second wording in FR and RM). The Romansh templates were not reviewed by a native speaker (exploratory). The language effect could partly reflect template wording; a variant without the abstention cue is future work.
- Gold comes from Wikidata, so its errors propagate. Mitigations: the rule-based checks, the 25-landmark live re-check and the AI-assisted re-read of every core Apertus grader disagreement against the gold. The photo screening also found one Wikidata main photo of the wrong building (removed).
- Fame is proxied by Wikipedia article counts. Low accuracy everywhere limits the power of fame contrasts (floor).
- Neither automatic grader is perfect. For the judge's errors, see Results 7. The code grader (second rater) misses fine-grained contradictions: Romansh "house of X" phrases, a denomination the gold does not state, name-shaped phrases built from common nouns ("Tour de l'Horloge" for another tower), and descriptive museum names.
- Corrections to the code grader and question validity were made after seeing answers; each is documented with its effect in `CHANGELOG.md` and pinned by regression tests. A second independent review read every answer the code grader marked incorrect and moved 109 (70B) and 356 (8B) labels, mostly refusals phrased in ways the patterns did not know; a third read 831 labels, and its fixes moved 14 (70B) and 29 (8B) more. Every changed label was read individually; these reviews were AI-assisted, not a human audit.
- With one annotator, the human audit cannot report agreement between people. The annotator's labels depart from the written rules in places (declines marked incorrect), which lowers the judge–human agreement.
- The photo screening was done by AI (two systems), with the team lead deciding edge cases; no person reviewed every shipped photo. It removed 13 landmarks; of the 4 removed for privacy, 3 were busy town-centre scenes, also among the easiest to recognise, so the removal took away 3 of the 70B's 9 correct place answers from photos (see H3).

Lessons Learnt and Recommendations

- For Apertus: its visual knowledge of Swiss landmarks lags a comparable open model (multiple choice 35 % vs 61 % for Gemma-4-31B). Its strength is calibration (declining instead of guessing in French and Italian), and Romansh breaks it. Calibration training should include Romansh and German, and objects that look recognisable but are unknown; retrieval (e.g. a KGS/Wikidata lookup) would help recognition more than more text.
- For evaluators: score abstention separately from accuracy and compare languages on identical images; check hedged answers ("not sure, but it seems to be a church") — they are abstentions, not errors.

Reproducibility

`make run` (Docker) on a clean checkout builds the image, downloads the pinned dataset revision from Hugging Face and runs `verify` on the reference results, Apertus 70B (schema, image hashes, coverage, the code grader reproduces every stored code label, final-label rule, recomputed scores equal the expected scores). If `LLM_*` is set, it also evaluates that model on a stratified sample (`FULL=1` for all items). Hardware: any CPU; runtime ≈ 1 min for verification, under an hour for a full 70B run at 4 parallel requests. Seeds: selection 20261013, sampling 7, bootstrap 20261015, stability 20261016, audit 20261017. Dataset revision `26d86b35d1a6` (pinned in `config.json`; the first release, 3,412 core items; the re-release with the extensions is pending); code: the `main` branch, tag `submission`.

License

Code: Apache-2.0. Dataset: CDLA-Permissive-2.0 (photos under their own CC licences, listed per item). Report: CC-BY-4.0.

References

SimpleQA (Wei et al., 2024); Wikidata; Swiss Inventory of Cultural Property of National and Regional Significance (KGS); BFS Swiss official commune register; Apertus technical report (Swiss AI Initiative, 2025).